

Electroons Are Awesome!

I am a new subscriber to EFY magazine. The cartoons published under 'Electroons' section are awesome! The sense of humour is really appreciable. The 'Arduino UNO-based Shadow Alarm' construction project published in February issue is also good. I request you to publish a buyers' guide on soldering stations.

Tanmay Dasgupta
Bilaspur, Chhattisgarh

EFY: Thank you for such a wonderful feedback. We will definitely forward your request to the concerned.

GSM-based Borewell Water Pump

I want to assemble the 'GSM-based Borewell Water Pump' construction project published in May 2012 issue. Please tell me how to burn the program into the AT89C51 microcontroller. What hardware and software should I use for the purpose?

Jaideep Mehendale
Through e-mail

The author Gurunath Reddy replies:

I have used an ATME1 series microcontroller burner—purchased from NSK Electronics, Bengaluru—to program the AT89C51 microcontroller. The source code was compiled using µVision3 IDE, and the hex file was loaded into the microcontroller using Proload software.

Make Your Own Web Server

I liked the 'Make Your Own Web Server' DIY article published in January issue. I have tried this project and built a Web server without any problem. I thank EFY and the author of this project. I also want to know how I can send 'live' videos from a webcam to my webpage.

Swami Nath Bharti
Through e-mail

The author Aadeeb Raza replies:

Thanks to the reader for the appreciation! Regarding the query, it is not easy to explain in a few steps. I will try to write another article on the same topic for EFY. For now, one can use www.

From Facebook: EFY's Electronics Design Community

I am looking for a long-range RF transmitter and receiver. Please advise if anyone has used it. I need to use it inside the house with walls between the transmitter and the receiver.

Vinay Chaddha

What range are you expecting? Nowadays RF transmitters and receivers are available that can work up to a range of 1 km. If you are ready to spend about Rs 5000, go for XBEE pro series. RF transceivers with 1km range that cost as little as Rs 1000 are also available.

Sumit Vaish

A Zigbee-based system can be used to interface through serial communication.

Nishant Mittal

Xbee pro can be used. If you want point-to-point/multipoint communication, series 1 should serve the purpose. For mesh topology you can use series 2.

Rubal Chib

You can use WiFi technology with star topology. GPRS technology with traditional modem communication can also be used.

Sarangale Rahul

ustream.tv website to upload the live video from a webcam. For more information, one may visit the link 'https://www.ustream.tv/ustream'

Cellphone-based Remote Controller for Water Pump

In January issue, two useful projects have been published in Circuit Ideas section: 'Cellphone-based Remote Controller for Water Pump' and 'Remote Control Using Wireless Door Bell.'

In this context, I would like to inform EFY that in 2009 I had developed two similar types of remote control systems for my company—one mobile operated and the other wireless calling bell operated. RDCIS (R&D wing of SAIL) has applied for patent on behalf of my company with me as the inventor for these two innovations in 2010.

The circuits and purpose published in EFY are similar to my remote control systems.

Gautam Kumar Mandal
Assistant General Manager
Industrial Engineering Deptt
Durgapur Steel Plant, SAIL

Smart Solar Charger

In the 'Smart Solar Charger' construction project published in January issue, it is mentioned that the circuit charges the battery in boost mode and trickle-charge mode. It is also mentioned that energy and power are calculated and displayed on the LCD display.

'Spot An Error' Award Winners

In the 'Blown-fuse Indicator for AC Load' circuit idea published in February issue, the value of R3 is printed as 10-ohms, 2W in the parts list. The correct value is 100 ohms, 2W.

Adil Khan

□ In the 'Low-Cost 5V Power Supply' circuit idea, in Fig. 2 both terminals of TL431 are printed as 'Cathode (K)'. The lower terminal should be 'Anode (A)'.

Somnath Biswas

ERRATA

In 'Low-Cost 5V Power Supply' circuit published in February issue, the current through the 2.2k resistor will be 1.55 mA. This is clearly not a micro-ampere current as mentioned in the text. The pin-out of TL431 shows cathode for both the leads.

Colin Mitchell
Through e-mail

In this context, I would like to comment that:

1. This charger cannot trickle-charge the battery as the battery is directly connected to the output of the solar panel (source) through a relay. The battery is always loaded, powering the entire circuit. Considering the battery voltage as 12V, IC 7805 regulator consumes a lot of power as the voltage difference is about 7V (dropout voltage). The relay too is powered by the battery, which